

CHIMERA II OS / CHIMERA 8192

IEEE-STYLE STANDARDS REFERENCE + RFC-STYLE EXPERIMENTAL PROFILE

Research / Experimental — NOT an IEEE Standard and NOT an IETF RFC

Scope. This document maps the supplied Chimera II / Chimera 8192 architecture to relevant IEEE standards and IETF RFCs, while defining an RFC-style interoperability profile for future research standardization. The Chimera-specific ISA, 8192-bit processor model, simulator, zero-copy interfaces and robotics architecture remain project-defined research specifications.

1. Architecture Baseline

The supplied specification defines a configurable software-defined/emulated processor ecosystem with R8192/C8192, 1024 × 8192-bit GPRs, 128 × 64-bit logical lanes, research-target 8192-bit data and 4096-bit address buses, microkernel services, zero-copy networking, cross-platform HALs, BrainNetwork simulation, robotics and distributed intelligence.

The BrainNetwork model explicitly represents virtual Chimera CPUs as computational nodes in a graph; a scheduler advances active nodes, queued messages carry signals, and SDL2/OpenGL/Dear ImGui expose execution and memory state.

2. IEEE References

[I1] IEEE Std 1003.1-2024. Portable Operating System Interface (POSIX) Base Specifications, Issue 8.

Chimera use: Chimera POSIX shell, utilities, system interfaces and application portability

<https://standards.ieee.org/ieee/1003.1/7700/>

[I2] IEEE Std 754-2019. IEEE Standard for Floating-Point Arithmetic.

Chimera use: Floating-point semantics for vector/tensor/AI execution

<https://standards.ieee.org/ieee/754/6210/>

[I3] IEEE Std 802.3-2022. IEEE Standard for Ethernet.

Chimera use: Ethernet interoperability for the NIC and zero-copy network stack

<https://standards.ieee.org/ieee/7003/10422/>

[I4] IEEE Std 1588-2019. Precision Clock Synchronization Protocol for Networked Measurement and Control Systems.

Chimera use: Distributed robotics, deterministic timing and swarm-node clock synchronization

<https://standards.ieee.org/ieee/1588/6825/>

[I5] IEEE 802.1Q family. Bridges and Bridged Networks / Virtual Bridged LANs.

Chimera use: Optional VLAN and bridged-network deployment

<https://standards.ieee.org/ieee/802.1Q/4410/>

3. IETF / RFC References

[R1] RFC 2119. Key words for use in RFCs to Indicate Requirement Levels.

Chimera use: MUST/SHOULD/MAY normative vocabulary

<https://www.rfc-editor.org/info/rfc2119/>

[R2] RFC 8174. Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words.

Chimera use: Clarifies uppercase BCP 14 requirement keywords

<https://www.rfc-editor.org/info/rfc8174/>

[R3] RFC 1122. Requirements for Internet Hosts — Communication Layers.

Chimera use: Host networking architecture and link/IP/transport requirements

<https://www.rfc-editor.org/info/rfc1122/>

[R4] RFC 791. Internet Protocol (IPv4).

Chimera use: IPv4 packet and Internet-layer interoperability

<https://www.rfc-editor.org/info/rfc791/>

[R5] RFC 8200. Internet Protocol, Version 6 (IPv6) Specification.

Chimera use: IPv6 packet and Internet-layer interoperability

<https://www.rfc-editor.org/info/rfc8200/>

[R6] RFC 768. User Datagram Protocol.

Chimera use: Low-overhead datagram transport for sensor/control traffic

<https://www.rfc-editor.org/info/rfc768/>

[R7] RFC 9293. Transmission Control Protocol (TCP).

Chimera use: Current consolidated TCP standards-track reference

<https://www.rfc-editor.org/rfc/rfc9293.pdf>

[R8] RFC 4861. Neighbor Discovery for IP version 6 (IPv6).

Chimera use: IPv6 neighbor/router discovery and reachability

<https://www.rfc-editor.org/info/rfc4861/>

[R9] RFC 4301. Security Architecture for the Internet Protocol.

Chimera use: IPsec security architecture reference

<https://www.rfc-editor.org/info/rfc4301/>

[R10] RFC 8446. The Transport Layer Security (TLS) Protocol Version 1.3.

Chimera use: TLS 1.3 reference; verify current successor/status before normative adoption

<https://www.rfc-editor.org/info/rfc8446/>

4. Standards Interoperability Matrix

Chimera subsystem	Reference	Use
POSIX subsystem	IEEE 1003.1-2024	API/shell portability
Floating point	IEEE 754-2019	Numerical semantics
Ethernet	IEEE 802.3-2022	MAC/PHY interoperability
Distributed timing	IEEE 1588-2019	PTP clock synchronization
VLAN/bridging	IEEE 802.1Q family	Network segmentation
Normative language	RFC 2119 + RFC 8174	Future RFC-style requirements
Host networking	RFC 1122	Layered network host behavior
IPv4	RFC 791	Internet layer
IPv6	RFC 8200	Internet layer
UDP	RFC 768	Datagram transport
TCP	RFC 9293	Reliable transport
IPv6 ND	RFC 4861	Neighbor/router discovery
IPsec	RFC 4301	Network security
TLS	RFC 8446 / successor	Session/application security

5. RFC-Style Experimental Profile

Status: Experimental / Informational. This section uses RFC conventions only; it has no IETF publication status and makes no IANA allocation.

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" are to be interpreted according to BCP 14 (RFC 2119, updated by RFC 8174). [cite]turn2search0[turn2search2]

5.1 Normative Chimera Requirements

R8192 ISA: SHOULD expose a stable, versioned 64-bit instruction encoding and opcode space.

C8192 ISA: MUST provide unambiguous variable-length packet boundaries.

8192-bit GPRs: Defines 1024 × 8192-bit architectural registers; physical implementations MAY bank/slice them.

128 × 64-bit lanes: Is an execution organization, not evidence of a particular silicon microarchitecture.

4096-bit address bus: Is a research target; software MUST distinguish architectural address width from host/emulator limits.

Zero-copy buffers: MUST use explicit ownership transitions so NIC, kernel and user space cannot concurrently mutate one buffer.

BrainNetwork: MUST treat virtual CPUs as computational graph nodes, not biological neurons.

Deterministic simulation: SHOULD support reproducible scheduling, message ordering and state replay.

Robotics: Hard-real-time control loops MUST be isolated from non-real-time visualization/cloud workloads.

Security: Secure boot, W^X , MMIO protection, DMA/IOMMU isolation and capability checks SHOULD be validated independently.

5.2 Networking Profile

For Internet interoperability, Chimera SHOULD use IEEE Ethernet at the link layer, IPv4/IPv6 at the Internet layer, and UDP/TCP at the transport layer. IEEE 802.3 supplies the Ethernet reference; RFC 1122 defines host communication-layer requirements; RFC 791 and RFC 8200 define IPv4/IPv6; RFC 768 defines UDP; RFC 9293 is the current consolidated TCP specification. [cite]turn0search8[turn2search1][turn1search2][turn1search4][turn1search0][turn1search35]

Zero-copy is a Chimera implementation optimization, not an Internet protocol requirement. Therefore the implementation MUST preserve external protocol semantics while allowing DMA rings, shared buffers, page pinning and descriptor ownership internally.

5.3 Distributed Robotics and Timing

IEEE 1588-2019 is the principal external timing reference for deterministic distributed robotics. It defines Precision Time Protocol for synchronizing clocks in networked measurement and control systems and supports sub-microsecond synchronization in suitable deployments. [cite]turn4search0]

Chimera-specific scheduler determinism, control deadlines, swarm state synchronization and replay semantics require a separate profile.

5.4 Security

The supplied architecture proposes secure boot, W^X, protected MMIO, DMA/IOMMU isolation, capability checks and constant-time cryptography. These are project requirements rather than direct requirements of one IEEE/RFC document. RFC 4301 provides the IPsec security architecture reference. RFC 8446 is a TLS 1.3 reference, but its current status should be checked before freezing a normative TLS version because it has been obsoleted by RFC 9846.

□cite□turn3search3□turn3search2□

6. IEEE-Style Bibliography

- [1] IEEE, “IEEE/Open Group Standard for Information Technology—Portable Operating System Interface (POSIX™) Base Specifications, Issue 8,” IEEE Std 1003.1-2024, 2024.
- [2] IEEE, “IEEE Standard for Floating-Point Arithmetic,” IEEE Std 754-2019, 2019.
- [3] IEEE, “IEEE Standard for Ethernet,” IEEE Std 802.3-2022, 2022.
- [4] IEEE, “IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems,” IEEE Std 1588-2019, 2019.
- [5] IEEE, “Bridges and Bridged Networks / Virtual Bridged LANs,” IEEE 802.1Q family.
- [6] J. Postel, “Internet Protocol,” RFC 791, IETF, Sep. 1981.
- [7] S. Deering and R. Hinden, “Internet Protocol, Version 6 (IPv6) Specification,” RFC 8200, IETF, Jul. 2017.
- [8] J. Postel, “User Datagram Protocol,” RFC 768, IETF, Aug. 1980.
- [9] W. Eddy, Ed., “Transmission Control Protocol (TCP),” RFC 9293, IETF, Aug. 2022.
- [10] R. Braden, Ed., “Requirements for Internet Hosts—Communication Layers,” RFC 1122, IETF, Oct. 1989.
- [11] T. Narten, E. Nordmark, W. Simpson, and H. Soliman, “Neighbor Discovery for IP version 6 (IPv6),” RFC 4861, IETF, Sep. 2007.
- [12] S. Kent and K. Seo, “Security Architecture for the Internet Protocol,” RFC 4301, IETF, Dec. 2005.
- [13] E. Rescorla, “The Transport Layer Security (TLS) Protocol Version 1.3,” RFC 8446, IETF, Aug. 2018.
- [14] S. Bradner, “Key words for use in RFCs to Indicate Requirement Levels,” BCP 14, RFC 2119, IETF, Mar. 1997.
- [15] B. Leiba, “Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words,” BCP 14, RFC 8174, IETF, May 2017.

7. Chimera-Specific vs External Standardized Components

- R8192 and C8192 instruction encodings, opcode definitions and execution semantics.
- 1024 × 8192-bit GPR architectural model and 128 × 64-bit execution-lane model.
- 8192-bit data-bus and 4096-bit address-bus research targets.
- TensorFS and Chimera-specific VFS integration.
- Zero-copy buffer ownership state machine and Chimera packet descriptors.
- BrainNetwork / Virtual ChimeraCPU graph execution and visualization model.
- Robotics HAL, swarm behavior, distributed learning and deterministic replay.
- Any claimed performance multiplier, silicon frequency, energy efficiency, or biological brain-emulation capability.

8. Recommended Standardization Roadmap

- 1 Freeze and version the R8192/C8192 ISA and publish machine-readable opcode definitions.
- 2 Define the ABI: endianness, alignment, calling convention, exception model, object format and relocations.
- 3 Define the zero-copy descriptor ABI and buffer ownership state machine.
- 4 Define deterministic scheduling, timestamping, replay and state-snapshot semantics.
- 5 Define a robotics timing profile using IEEE 1588 as the external clock reference.
- 6 Define the security profile and threat model, including secure boot, key lifecycle and DMA/IOMMU isolation.
- 7 Publish an Internet-Draft-style document before describing any text as an RFC.
- 8 Separate architectural targets, emulator results, FPGA measurements and silicon measurements in all performance claims.

9. Evidence Boundary

The source specification explicitly identifies the 8192-bit datapath, register file and accelerator blocks as research targets rather than existing silicon, and requires FPGA or silicon measurements for physical performance claims.

[\[filecite\]turn4file0L17-L23](#) The roadmap separates emulator, kernel, UEFI, compiler, networking, BrainNetwork, robotics and FPGA stages. [\[filecite\]turn4file2L134-L147](#)

This distinction should remain explicit in any future IEEE-style paper, technical report, or Internet-Draft.